

- 6.4 For this function, find a minimum sum-of-products solution, using the Quine-McCluskey method.

$$f(a, b, c, d) = \Sigma m(1, 3, 4, 5, 6, 7, 10, 12, 13) + \Sigma d(2, 9, 15)$$

- 6.5** Find all prime implicants of the following function and then find all minimum solutions using Petrick's method:

$$F(A, B, C, D) = \Sigma m(9, 12, 13, 15) + \Sigma d(1, 4, 5, 7, 8, 11, 14)$$

6.6 Using the method of map-entered variables, use four-variable maps to find a minimum sum-of-products expression for

(a) $F(A, B, C, D, E) = \Sigma m(0, 4, 5, 7, 9) + \Sigma d(6, 11) + E(m_1 + m_{15})$, where the m 's represent minterms of the variables A, B, C , and D .

(b) $Z(A, B, C, D, E, F, G) = \Sigma m(0, 3, 13, 15) + \Sigma d(1, 2, 7, 9, 14)$
 $+ E(m_6 + m_8) + Fm_{12} + Gm_5$

6.7 For each of the following functions, find all of the prime implicants using the Quine-McCluskey method.

(a) $f(a, b, c, d) = \Sigma m(0, 3, 4, 5, 7, 9, 11, 13)$

(b) $f(a, b, c, d) = \Sigma m(2, 4, 5, 6, 9, 10, 11, 12, 13, 15)$

6.8 Using a prime implicant chart, find *all* minimum sum-of-products solutions for each of the functions given in Problem 6.7.

(a) $f(a, b, \bar{c}, d) = \Sigma m(0, 3, 4, 5, 7, 9, 11, 13)$

(b) $f(a, b, c, d) = \Sigma m(2, 4, 5, 6, 9, 10, 11, 12, 13, 15)$

6.9 For each function, find a minimum sum-of-products solution using the Quine-McCluskey method.

(a) $f(a, b, c, d) = \Sigma m(2, 3, 4, 7, 9, 11, 12, 13, 14) + \Sigma d(1, 10, 15)$

(b) $f(a, b, c, d) = \Sigma m(0, 1, 5, 6, 8, 9, 11, 13) + \Sigma d(7, 10, 12)$

(c) $f(a, b, c, d) = \Sigma m(3, 4, 6, 7, 8, 9, 11, 13, 14) + \Sigma d(2, 5, 15)$

6.18 The following prime implicant table (chart) is for a four-variable function $f(A, B, C, D)$.

	2	3	7	9	11	13
-0-1		x		x	x	
-01-	x	x			x	
--11		x	x		x	
1--1				x	x	x

(a) Give the decimal representation for each of the prime implicants.

(b) List the maxterms of f .

(c) List the don't-cares of f , if any.

(d) Give the algebraic expression for each of the essential prime implicants.

- 6.19** Packages arrive at the stockroom and are delivered on carts to offices and laboratories by student employees. The carts and packages are various sizes and shapes. The students are paid according to the carts used. There are five carts and the pay for their use is
- Cart C1: \$2
 - Cart C2: \$1
 - Cart C3: \$4
 - Cart C4: \$2
 - Cart C5: \$2

On a particular day, seven packages arrive, and they can be delivered using the five carts as follows:

- C1 can be used for packages P1, P3, and P4.
- C2 can be used for packages P2, P5, and P6.
- C3 can be used for packages P1, P2, P5, P6, and P7.
- C4 can be used for packages P3, P6, and P7.
- C5 can be used for packages P2 and P4.

The stockroom manager wants the packages delivered at minimum cost. Using minimization techniques described in this unit, present a systematic procedure for finding the minimum cost solution.

6.23 Find all prime implicants of the following function, and then find all minimum solutions using Petrick's method:

$$F(A, B, C, D) \mid = \Sigma m(7, 12, 14, 15) + \Sigma d(1, 3, 5, 8, 10, 11, 13)$$